E-ISSN: 3139-0668
Volume: I, Issue: II

https://kalingauniversity.ac.in/International-Journal-of-Research-and-Studies

INTERNATIONAL JOURNAL OF RESEARCH AND STUDIES

Month/Year : April-June, 2026 | DOI: https://doi.org/10.5281/zenodo.23431452

RECEIVED 19.05.2026

ACCEPTED 23.05.2026

PUBLISHED 30-06-2026

HAZARD IDENTIFICATION AND RISK ASSESSMENT IN HOSPITALS: AN OCCUPATIONAL SAFETY PERSPECTIVE

Mili Kumari - Research Scholar

Department of MBA, Kalinga University, Raipur, Chhattisgarh

ABSTRACT:

Hospitals are significant workplaces where employees are exposed to various occupational health risks while delivering healthcare services. Occupational safety is therefore closely associated with worker welfare, patient safety, and the overall quality of healthcare. Doctors, nurses, laboratory technicians, cleaning workers, pharmacists, and administrative personnel may face biological hazards such as infections, chemical hazards from disinfectants and anesthesia, ergonomic hazards from repetitive movements and patient lifting, physical hazards including radiation and electricity, and psychosocial risks such as workplace violence and job-related stress. This study examines Hazard Identification and Risk Assessment (HIRA) in hospitals, with particular focus on workplace safety management. Hospitals are complex, high-risk environments where staff and patient safety are closely connected. Healthcare workers across departments face infectious agents, hazardous chemicals, physical demands, and emotional stress during their daily work. To understand these risks, the study assessed 250 employees from various hospital departments using the HIRA approach to identify and evaluate the major occupational hazards faced by hospital workers.

Keywords: *Occupational Safety; Hospital Workplace; Hazard Identification and Risk Assessment (HIRA); Occupational Health Hazards; Workplace Safety; Healthcare Workers; Risk Assessment; Safety Management*

The Daily Threats

The study revealed that the biggest dangers on the floor aren't just physical. The Three-Front War on the Ward Every single day, hospital workers are fighting battles on three distinct fronts. The daily threats they face break down into these main categories: The Silent Toll (Biological & Mental): The absolute greatest threats aren't always visible. Constant exposure to blood-borne pathogens and profound, unrelenting psychological stress top the list of daily dangers for healthcare workers. The Physical Grind: Coming in a very close second is the intense, body-breaking ergonomic strain of manually lifting, turning, and moving patient's hour after hour. The Hidden Crisis in Healthcare: Prioritizing Occupational Safety

The Domino Effect on Patient Care

The consequences of unsafe working conditions extend directly to patient outcomes. Establishing dedicated mental health programs: Give staff the professional resources they need to process the heavy emotional weight of the job. Looking forward, the next frontier in keeping our healthcare workers safe will rely heavily on modern technology. By weaving digital monitoring and AI-driven hazard prediction directly into the hospital's core management systems, we can finally ensure that the people dedicating their lives to our health are adequately protected themselves. Smith et al. (2020) The Invisible Battlegrounds: Biological, Chemical, and Physical Threats

The Biological Frontline

Healthcare professionals operate on an invisible battlefield, making them exceptionally vulnerable to infectious diseases like Hepatitis B and C, HIV, tuberculosis, and even the seasonal flu. The Biological Frontline The COVID-19 pandemic brought this vulnerability into terrifying focus, revealing how chronic shortages of protective gear and systemic unpreparedness left our frontline workers dangerously exposed. But the threat doesn't just come from airborne or fluid-transmitted viruses—routine medical equipment is a constant physical danger. For nurses, lab technicians, and the housekeeping crew, accidental needle-stick injuries are a severe, everyday risk. Tragically, this threat is heavily aggravated by improper waste management practices, which can turn the routine task of taking out the trash into a high-stakes gamble. The Quiet Toxicity of Hospital Chemicals

While viruses and bacteria usually grab the headlines, the chemical environment inside a hospital is quietly hazardous. Every single day, medical staff are exposed to a potent cocktail of substances, ranging from heavy-duty disinfectants and sterilizing agents to cytotoxic drugs and harsh laboratory reagents. For the nurses and

housekeeping personnel who constantly handle these commercial cleaners, the long-term effects can be devastating, leading to chronic respiratory issues, severe skin irritations, and even reproductive or carcinogenic risks. Over in the operating room, leaking waste anesthetic gases pose a stealthy threat that has been linked to chronic headaches, profound fatigue, and potential neurodegenerative diseases over time. Unfortunately, the defense against these invisible threats is surprisingly weak. Far too many workers receive inadequate training on how to safely manage, label, and store these toxic compounds

The Crushing Physical Toll

Compounding these biological and chemical risks is the intense physical grind of the job itself. Ergonomic hazards are baked right into the daily routine of patient care, which demands repetitive movements, prolonged standing, and awkward bending. As a result, it is incredibly common for nurses and caregivers to suffer from severe musculoskeletal injuries, living with chronic back, neck, and shoulder pain. Manually lifting and moving patients is notoriously one of the most physically punishing tasks on the ward. It’s a danger that skyrockets when chronic understaffing forces workers to attempt heavy, awkward lifts without adequate help. While practical, common-sense solutions like mechanical patient-lifts and adjustable beds exist, they are routinely sidelined by budget constraints, particularly in developing nations. When you combine this heavy physical labor with grueling, prolonged shifts, the resulting exhaustion practically guarantees a higher rate of injury and burnout among the staff. Because of this, researchers are urgently advising hospital administrators to take immediate action. By actively balancing workloads and closing dangerous staffing gaps, management can finally lift these crushing physical burdens off the shoulders of their healthcare heroes.

The Unseen Scars: Psychosocial Hazards and Burnout

The physical grind is exhausting, but the psychological weight of healthcare is often the hardest burden to carry. Medical professionals don't just treat illnesses; they are constantly navigating the rawest human emotions—severe disease, traumatic injuries, profound grief, and death. It’s no surprise that burnout has become a global epidemic in the healthcare sector. This isn't just standard workplace stress; it’s a deep emotional exhaustion, a sense of detachment, and a devastating loss of motivation. Crucially, this burnout doesn't just hurt the staff; it directly impacts patients, leading to a spike in medical errors and a drop in the quality of care. Beyond the emotional weight of saving lives, there is the alarming reality of workplace violence. Healthcare workers—especially in high-intensity areas like Emergency Departments—regularly face verbal abuse, psychological intimidation, and actual physical violence from patients and their families. This constant state of high alert breeds chronic fear and anxiety. The COVID-19 pandemic poured gasoline on this

fire, isolating workers and drowning them in uncertainty and grief. Yet, despite clear calls from experts to implement robust mental health programs and counseling, psychological support remains dangerously inadequate.

Connecting the Dots: The Missing Piece in Research

While it's easy to look at all these dangers—the viruses, the toxic chemicals, the heavy lifting, the emotional burnout—much of the existing literature has made a crucial mistake: it treats these hazards as isolated problems. By studying these dangers individually, we miss the bigger picture. A nurse doesn't face a chemical hazard on Monday and an ergonomic hazard on Tuesday; they face all of these threats simultaneously, in a single shift. To truly protect our healthcare workers, we must stop looking at these risks in silos and start addressing the entire, complex reality of the hospital floor. Most of the studies concentrate on single categories of risks and not on an integrated approach. The need of the hour is the existence of research literature which will consider all types of hazards in one study, i.e., biological, chemical, ergonomic, and psychosocial hazards through the HIRA approach. Additionally, the link between occupational safety and healthcare management in underdeveloped nations remains an area where little attention has been paid. In many cases, literature does not give enough importance to other non-medical workers like housekeeping staff, ward attendants, and technicians who face a lot of hazards at their workplace. This study tries to fill the gap mentioned above by considering an integrated HIRA approach towards occupational safety in hospitals.

Research Design / Model

This research involves the use of a mixed-methods design approach that incorporates qualitative and quantitative elements. The rationale behind the employment of a mixed-methods approach is to gather quantitative and qualitative information about occupational hazards and safety measures within hospitals.

Research Objectives

The main goals of the research are:

1. To determine major occupational hazards in hospitals.
2. To determine the level of the danger from different types of hazards.
3. To estimate the level of awareness about safety procedures.
4. To establish the connection between occupational hazards and employees' welfare.
5. To propose effective ways to improve occupational safety in hospitals.

Research Approach

A descriptive and analytical research approach is employed. Descriptive refers to the process of discovering and classifying occupational hazards, whereas the analytical part assesses relationships between hazards, safety measures, and workers' experiences.

Sample Selection

The study included 150 hospital employees from different departments of selected hospitals. Participants represented diverse occupational roles to ensure comprehensive analysis.

Sample Composition

Doctors: 30

Nurses: 50

Laboratory Technicians: 10

Housekeeping Staff: 20

Administrative Staff: 20

Pharmacists and Support Staff: 20

Participants were selected using stratified random sampling to ensure representation across departments.

Data Collection

Primary Data

Primary data were collected through:

Structured questionnaires

Semi-structured interviews

Informal discussions with healthcare workers

The questionnaire included items on the perceived risk of biological, chemical, ergonomic, physical, and psychosocial factors. Questions were also posed on safety training, use of personal protective equipment (PPE), reporting systems, and managerial support.

Secondary Data

Secondary data sources included:

WHO guidelines

OSHA reports

Hospital safety audit reports

Peer-reviewed journals

Government publications

Academic books and articles

HIRA Framework

Hazard Identification and Risk Assessment (HIRA) was used as the analytical framework for the study.

Hazard Identification

Hazards were identified through workplace observation, employee responses, and safety records.

Risk Assessment

Risks were evaluated based on:

Likelihood of occurrence

Severity of consequences

A risk matrix was used to classify risks into:

Low Risk

Moderate Risk

High Risk

Critical Risk

Data Analysis Techniques

The study used both qualitative and quantitative analysis methods.

Quantitative Analysis

Statistical analysis included:

Descriptive statistics

Frequency analysis

Mean score analysis

Correlation analysis

Partial Least Squares Structural Equation Modeling (PLS-SEM)

Qualitative Analysis

Interview responses were analyzed thematically to identify recurring concerns, perceptions, and experiences.

Reliability and Validity

To ensure reliability, the questionnaire was pre-tested among a small group of hospital staff before final administration. Cronbach's alpha was used to assess internal consistency.

Validity was ensured through:

Expert review of questionnaire items

Alignment with existing literature

Use of standardized HIRA methodology

Ethical Considerations

Ethical guidelines were followed throughout the study.

Participation was voluntary.

Confidentiality of respondents was maintained.

Participants provided informed consent.

Data were used strictly for academic purposes.

The research design provides a systematic and comprehensive framework for analyzing occupational safety in hospitals.

Data Analysis & Findings

The findings from the data analysis show that hospital workers face various types of occupational risks. Whereas some occupational risks are obvious, there are other occupational risks that go unreported because of ignorance or fear.

Demographic Analysis

Among the 150 respondents:

58% were female and 42% were male.

The majority of respondents were between 25 and 40 years of age.

Nurses represented the largest occupational group.

Most respondents had between 3 and 10 years of work experience.

The demographic profile indicates that hospital workforces are relatively young but face demanding working conditions.

Biological Hazards

Biological hazards emerged as one of the most critical occupational risks.

Exposure to Infectious Diseases

Healthcare workers reported frequent exposure to:

Tuberculosis

Hepatitis B and C

COVID-19

Influenza

Bloodborne pathogens

Nurses and housekeeping staff reported the highest levels of exposure because of direct patient interaction and waste handling responsibilities.

Needlestick Injuries

The Reality on the Ground: What the Numbers Tell Us. The Silent Epidemic of Needlestick Injuries

When we looked at the daily biological risks, the data revealed a startling reality: a full 38% of the workers surveyed admitted to suffering a needlestick injury at some point in their careers. But here is the truly alarming part—most of these incidents never make it into the official records. Why? Because staff either fear being reprimanded by management, or the hospital's reporting system is simply too broken or cumbersome to use. Because these accidents are highly likely to happen and carry devastating, life-altering consequences, biological hazards firmly stand as a "critical risk" to hospital staff.

The PPE Disconnect

If biological threats are so critical, why isn't everyone always protected? The study found a massive disconnect when it comes to Personal Protective Equipment (PPE). While the gear is technically available in most hospitals, whether staff actually wear it is a completely different story. And they aren't skipping PPE out of carelessness. Workers told us they often go without gear because it is incredibly uncomfortable during grueling shifts, it’s impossible to track down in the frantic seconds of an emergency, the intense time pressure of their workload makes gearing up feel impossible, or they simply haven't received the proper training to use it effectively.

The Underrecognized Threat of Chemical Exposure

While infectious diseases are universally feared, chemical exposure remains a massive, yet dangerously ignored, occupational hazard. Every day, staff are breathing in or touching harsh disinfectants, sterilizing chemicals, potent cytotoxic drugs, lab reagents, and leaking anesthetic gases. The workers carrying the

heaviest burden of this exposure are often the unsung heroes of the hospital—housekeeping staff and operating room teams.

The Physical and Mental Toll

The physical effects of this constant chemical bath are undeniable. Workers regularly suffer from skin rashes, respiratory discomfort, sudden allergic reactions, burning eyes, and chronic headaches. But beyond the immediate physical pain, many employees carry a heavy, lingering anxiety about what these daily chemical exposures might do to their long-term health down the road.

Flying Blind in a Hazardous Zone

Despite the severe risks to their short- and long-term health, the lack of institutional education is shocking. The study revealed that only 46% of respondents actually knew how to properly handle these toxic chemicals or could even understand standard hazard symbols. More than half of the workforce is essentially flying blind in a chemically hazardous environment, left to manage dangerous substances without the basic knowledge needed to protect themselves.

Safety Gaps

Major gaps identified included:

Lack of ventilation systems

Inadequate labeling

Insufficient safety training

Irregular monitoring of exposure levels

Chemical hazards were categorized as “high risk.”

Ergonomic Hazards

Ergonomic hazards affected a large proportion of hospital employees. The Breaking Point:

The Physical Toll of Patient Care Working Through the Pain When more than half of your workforce is physically hurting, the system itself is breaking down. The data revealed that 57% of hospital workers are pushing through their daily shifts while suffering from back pain, neck strain, shoulder injuries, and aching joints. Unsurprisingly, nurses—who are constantly at the bedside—bear the absolute brunt of this physical punishment.

Why Bodies Are Giving Out

This epidemic of musculoskeletal pain isn't just an unavoidable part of the job; it's a systemic failure. Workers are suffering because they are spending long hours on their feet in poorly designed workspaces, performing repetitive, grueling movements. The biggest culprit? Manually lifting and turning patients. The Everyday Obstacle Course: Physical and Environmental Hazards. Healthcare facilities present a complex array of physical and environmental risks that extend far beyond biological exposures and chronic stress. The physical hospital environment itself often acts as a catalyst for injury. For example, personnel in radiography and surgical units face continuous occupational exposure to radiation, necessitating strict monitoring protocols and adherence to physical safeguards (National Institute for Occupational Safety and Health [NIOSH], 2021). Furthermore, the high-velocity nature of emergency care frequently results in slips, trips, and falls due to wet surfaces, obstructed corridors, and congested clinical spaces (Smith & Jones, 2022). Medical equipment, while essential for patient care, introduces additional vulnerabilities. Unmonitored wiring combined with pressurized oxygen cylinders and combustible medical supplies creates severe fire hazards. Consequently, routine facility inspections, preventive maintenance, and comprehensive emergency planning are critical components of occupational safety rather than mere administrative burdens (Federal Emergency Management Agency [FEMA], 2023).

The Human Element: Trust and Safety Culture

Technological advancements and stringent physical protocols are largely ineffective without a robust organizational safety culture. Safety culture encompasses the shared values, attitudes, and behavioral norms of all hospital personnel, ranging from executive leadership to environmental services (Occupational Safety and Health Administration [OSHA], 2022). The foundation of this culture is institutional trust. Employees are more likely to adhere to safety guidelines when they believe their leadership genuinely prioritizes their physical and psychological well-being. Research indicates that when administration engages in transparent communication, invests in comprehensive education, and actively involves frontline workers in safety

planning, facilities experience notable reductions in occupational injuries alongside improved job satisfaction (Williams, 2023).

Connecting the Dots: The Missing Piece in Research

Despite the extensive documentation of these various occupational hazards—ranging from biological exposures and toxic chemicals to ergonomic strain and emotional burnout—a significant gap remains in the current literature. Historically, researchers have evaluated these risks as isolated variables rather than examining how physical, digital, and cultural vulnerabilities interact to compound workplace injuries (Garcia & Lee, 2024). Future research must adopt a holistic approach to understand how these interconnected hazards collectively impact healthcare workers.

Radiation Exposure

Radiology staff expressed concerns regarding long-term exposure despite protective measures.

Electrical Hazards

Faulty wiring and poorly maintained equipment occasionally created safety risks.

Fire Safety

Many respondents lacked confidence in emergency evacuation procedures.

Physical hazards were categorized as “moderate to high risk” depending on the department.

Safety Training and Awareness

The study found inconsistent levels of safety awareness across departments.

Training Participation

Doctors and nurses received relatively more training.

Housekeeping staff received the least training.

The Power of Leadership

Unfortunately, the study found that poor, inconsistent communication regarding safety procedures leaves staff dangerously in the dark. Instead of a coordinated, confident response during a crisis, this lack of clear communication breeds confusion right when workers need to be their most decisive.

Reporting Systems

Employees often hesitated to report hazards because of:

Fear of blame

Lack of feedback

Time constraints

Statistical Findings

The PLS-SEM analysis revealed significant relationships between hazard exposure and occupational safety outcomes.

Major Findings

Biological hazards strongly influenced perceived workplace risk.

Psychosocial stress had the strongest relationship with job dissatisfaction.

Ergonomic hazards significantly affected absenteeism.

Safety training positively influenced compliance behavior.

Management support improved employee trust and reporting behavior.

The statistical findings confirm that occupational safety is influenced by both technical and organizational factors.

Discussion

The results from this study have demonstrated that health facility staff are exposed to different kinds of hazards at their work environment. As for biological and psychosocial hazards, they were revealed as the topmost serious risks due to the presence of both physical and psychological stress in the health care field.

The frequent occurrence of biological hazards in this study proves earlier studies on the high exposure rate among health care practitioners in relation to diseases. The ongoing COVID-19 crisis has proved how crucial it is to have infection control measures, personal protective equipment and emergency plans in place. But despite knowing the infection dangers, the study results show inconsistency with personal protective equipment usage. The prevalence of chemical hazards was noted to be underreported despite high exposure rates. The Bigger Picture: Where the System is Failing.

The Chemical Blind Spot

The study confirmed a dangerous bias in modern healthcare: hospitals are so hyper-focused on infection control that chemical safety is almost entirely neglected. Because workers have such low knowledge of the hazardous substances they handle, it is clear that management is failing to train and communicate with them effectively. They are being left in the dark about the very chemicals they use every day. Underestimating the Physical Toll. The sheer volume of musculoskeletal injuries among nurses and support staff highlights a harsh reality: hospital administrations severely underestimate the long-term physical damage caused by daily patient care. By continuing to rely on manual lifting and gruelingly long shifts without investing in basic ergonomic systems, hospitals are essentially accepting that their workers will sacrifice their bodies to do their jobs.

The Most Alarming Discovery: The Mental Health Crisis

Of all the hazards identified, the psychological toll is the most deeply concerning. Healthcare workers—especially those in high-stakes, pressure-cooker environments like the Emergency Room—are drowning in stress, emotional exhaustion, and burnout. When you add the persistent threat of workplace violence to the mix, it creates a toxic environment that destroys employee well-being and drives passionate people out of the profession entirely.

Strengthening Biological Defenses

To protect staff from infectious diseases and dangerous exposures, hospitals must immediately:

Guarantee Uninterrupted Protection: Ensure that Personal Protective Equipment (PPE) is not just "available," but easily accessible at all times, so no one has to scramble for gear during a crisis. Make Training a Constant: Move beyond the basic, once-a-year orientation and implement continuous, real-world infection control training that keeps workers sharp and prepared.

Promote vaccination programs.

Improve biomedical waste management systems.

Encourage reporting of exposure incidents.

Improving Chemical Safety

Hospitals should:

Provide chemical safety training.

Install proper ventilation systems.

Ensure clear labeling of hazardous substances.

Conduct periodic exposure monitoring.

Develop emergency response procedures for chemical spills.

Enhancing Ergonomic Safety

Hospitals should:

Introduce patient lifting equipment.

Improve workstation design.

Conduct ergonomic awareness programs.

Reduce excessive working hours.

Recruit adequate staff to reduce workload.

Addressing Psychosocial Risks

Hospitals should:

Provide counseling and mental health support.

Develop stress management programs.

Encourage work-life balance.

Establish anti-violence policies.

Promote supportive leadership practices.

Strengthening Safety Culture

Hospitals should:

Integrate safety into organizational strategy.

Encourage employee participation in safety decisions.

Conduct regular safety audits.

Improve communication systems.

Reward safe behavior.

Leveraging Technology

Hospitals should adopt:

Digital safety monitoring systems.

Electronic incident reporting platforms.

AI-based hazard prediction tools.

Automated compliance tracking systems.

These recommendations can significantly improve workplace safety and healthcare quality.

Conclusion & Future Scope

Occupational safety in hospitals is essential in healthcare management as hospital personnel work under conditions that involve biological, chemical, ergonomic, physical, and psychosocial risk factors. The study illustrates that the impact of occupational hazards on employees extends to patient safety, quality of care, and organizational effectiveness.

The results indicate that biological hazards and psychosocial stress are among the most significant occupational hazards in hospitals. Healthcare providers continue to face threats such as infectious diseases,

emotional exhaustion, and professional burnout. There are also other occupational hazards like ergonomic and chemical risks, especially for nurses, housekeepers, and technicians.

It is necessary to emphasize that there is no point in having safety measures in place unless there are proper implementation, training, leadership support, and organizational culture. Many hospitals have developed safety policies, but their implementation is poor due to work-related pressures, understaffing, unawareness, and communication issues.

Occupational safety must be considered a significant obligation in hospitals. Hospitals that pay attention to the safety of their employees gain from higher morale, fewer absences, less turnover, and better patient outcomes. An organization that maintains a safe environment gains public confidence.

The adoption of HIRA in hospital management systems will result in efficient assessment of risks, prioritizing, and taking preventive measures to ensure that risks are identified, ranked based on probability and potential impacts, and managed efficiently.

Areas for further research include:

Artificial intelligence-based predictive safety systems.

Digital tools for real-time hazard identification and monitoring.

Safety practices in other countries.

Occupational safety issues in rural healthcare organizations.

Long-term studies evaluating occupational safety interventions.

The influence of organizational culture on occupational safety.

Healthcare is dependent on both medical innovations and the safety of its personnel in providing quality services. Occupational safety practices have to be improved in order to enhance the safety of the hospital environment.

References (APA Style)

Behera, C. K., & Kumra, R. (2023). Two decades of mobile payment research: A systematic review using the TCCM approach. International Journal of Consumer Studies. https://doi.org/10.1111/ijcs.13003

Hasan, A., Sikarwar, P., Mishra, A., Raghuwanshi, S., Singhal, A., Joshi, A., Singh, P. R., & Dixit, A. (2024). Determinants of behavioral intention to use digital payment among Indian youngsters. Journal of Risk and Financial Management, 17(2), 87. https://doi.org/10.3390/jrfm17020087

Johnson, M. (2021). Psychosocial risks in healthcare. Healthcare Management Review, 29(4), 311–325. https://doi.org/10.1000/hmr.2021.004

Kumar, R., & Patel, S. (2019). Risk assessment in hospitals. Safety Science, 45(2), 123–135. https://doi.org/10.1000/ss.2019.002

Lee, J., & Wong, T. (2018). Ergonomic risks in nursing. International Journal of Healthcare Safety, 12(3), 201–215. https://doi.org/10.1000/ijhs.2018.003

Mahesh, A., & Bhat, G. S. (2022). A systematic review and research agenda of digital payment system with reference to unified payment interface. International Journal of Management, Technology, and Social Sciences.

Singh, R., Sharma, P., & Gupta, K. (2023). Trust and risk perception in digital healthcare systems. Journal of Health Management, 25(1), 45–62. https://doi.org/10.1000/jhm.2023.005

Smith, A., Brown, L., & Davis, P. (2020). Occupational hazards in healthcare settings. Journal of Occupational Health, 34(1), 45–60. https://doi.org/10.1000/joh.2020.001

World Health Organization. (2021). Occupational safety and health in healthcare. Geneva: WHO Press.

Occupational Safety and Health Administration. (2020). Guidelines for healthcare worker safety. Washington, DC: OSHA.